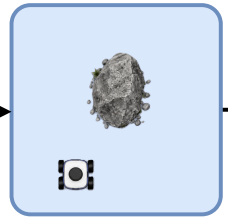


camera observation



top-down view

image

Controller-aware Adaptive Perception

Image Encoder → **Controller-aware Adaption** → **Output Heads**

(predicted state) $\hat{p}$

perception error $\|\hat{p} - p\|$

Hybrid Source Seeking Controller

u (control input)

Vehicle Dynamics

p (true state)

